

AI HORIZON SOLUTION CHALLENGE 2026 • NSW AUTOMATION

AI Dispensing Defect Detective — Troubleshooting Report

Tagline: “Helping Manufacturers Identify Dispensing Problems Faster with AI” | Dispense Lens Diagnostic Case Report

Case Identifier: CASE-2026-NSW-DEMO-01 | Report Revision: 1 | Case Created: 2026-09-22 15:00:34 UTC

1. Case Identity & Process Context

Case ID	CASE-2026-NSW-DEMO-01	Current Revision	1
Defect Category	D03_INCONSISTENT_SIZE	Issue Condition	UNRESOLVED
Defect Name	Inconsistent Dispensing Volume	Created Timestamp	2026-09-22 15:00:34 UTC
Material	epoxy_adhesive	Method	piezo_jetting
Problem Description	Dispensed epoxy dot diameter fluctuates ±18% across board positions after recent cartridge refill.	Machine Context	{'line': 'SMT-Bay-2', 'nozzle_gauge': '27G', 'pressure_bar': 1.8}

Dispensing Quality Assessment (NSW Bonus Challenge 2) — Overall Quality Score: 78 / 100

Quality Metric	Rating	Score	Status Assessment	Engineering Details
Shape Consistency	★★★★■ (4/5)	82%	Good Symmetry	Droplet circularity meets baseline specification with minor perimeter variation.
Size Consistency	★★★██ (3/5)	58%	Variable Diameter	Shot-to-shot dot volume fluctuates outside optimal tolerance band (±18%).
Dispensing Position	★★★★★ (5/5)	95%	Precise Alignment	Target XY pad alignment remains centered with near-zero positional skew.
Defect Risk	★★████ (2/5)	35%	Low-Moderate Risk	Mechanical bond integrity maintained; potential cosmetic and under-fill risk.

Quality Assessment Rationale: Good droplet geometry and target position, but volumetric repeatability requires parameter or fluid stabilization.

2. Current Outcome Summary

Current Condition	UNRESOLVED	Revision Basis	Revision 1
Confirmed Root Causes	None confirmed	Is Resolved	NO (Unresolved / In Progress)

3. Current Diagnosis Snapshot (Analysis Revision 1)

Diagnostic Explanation: Air bubbles are ranked as the highest possible cause (85% likelihood) because dispensing volume changes occasionally rather than continuously, and a recent syringe material refill was reported.

#	Cause ID	Cause Name	Score	Conclusion	Status
1	air_bubbles	Air Bubble in Syringe / Supply Path	85.0	SUSPECTED	Candidate
2	nozzle_restriction	Partial Nozzle Restriction	68.0	SUSPECTED	Candidate

Evaluated Evidence - Air Bubble in Syringe / Supply Path (air_bubbles):

Relation	Strength	Source	Score	Observation / Explanation Details
SUPPORTS	STRONG	USER	+40.0	Intermittent volume variation strongly suggests air bubbles moving through syringe. [obs: obs-3]

SUPPORTS	MODERATE	USER	+25.0	Recent material refill introduces high probability of air entrapment. [obs: obs-4]
----------	----------	------	-------	--

Evaluated Evidence - Partial Nozzle Restriction (nozzle_restriction):
No evaluated evidence recorded for this cause.

Recommended Next Question: [Q01] Does the problem happen continuously or only after prolonged operation? [Options: Continuously, After prolonged operation, Randomly] - *Differentiate air bubble entrapment from thermal viscosity thinning*
Recommended Next Troubleshooting Check: [ACT01] Inspect Syringe Column for Air Entrapment - Backlight syringe barrel and check for voids near luer lock.

AI Learning Database Insight (NSW Bonus Challenge 3)

AI Learning Insight: "Similar problems occurred 12 times previously. In 8 cases, the main cause was air trapped inside the syringe."
Historical Verified Resolution: Purged trapped air bubble from syringe luer-lock fluid path and re-zeroed dispensing back-pressure. Nominal dot volume restored.

4. Technician Question-Answer History - Question Answers (1)

#	Question ID	Answer Value / Text	Source	Rev	Timestamp
1	Q01	occasionally (Fluctuates occasionally during line run)	USER	1	2026-09-22 15:00:34 UTC

5. Troubleshooting-Check History - Troubleshooting Checks (1)

#	Check ID	Execution	Finding / Outcome	Rev	Timestamp
1	ACT01	COMPLETED	SUPPORTS: air_bubbles_observed (Air microbubbles observed in clear syringe luer fitting)	1	2026-09-22 15:00:34 UTC

6. Cause-Confirmation History - Cause Confirmations (0)

None recorded.

7. Issue Lifecycle History - Lifecycle Events (0)

None recorded.